1 | Experiential Learning Component

1.1 | UNIT IV : Experiential Learning - 12

[Level-1: 2Q, Level-2: 2Q]

1. **LEVEL - 1 : Implement Hash Table using Open Addressing (Linear Probing)**: Write a program to implement a hash table using linear probing for collision resolution.
   - **Input**: Insert keys into the hash table.
   - **Output**: Display the hash table after each insertion.
   - **Hint**: Use a hash function to determine the position in the table and probe sequentially to resolve collisions.
2. **LEVEL - 2 : Hash Table Insertion using Chaining**: Write a program to implement a hash table using chaining (linked lists) to resolve collisions.
   - **Input**: Insert keys into the hash table.
   - **Output**: Display the hash table with chains after each insertion.
   - **Hint**: Use a hash function to determine the position in the table. If a collision occurs, append the key to a linked list at that position.
3. **LEVEL - 1 : Search in a Hash Table**: Write a function to search for a given key in the hash table.
   - **Input**: A key to search for in the hash table.
   - **Output**: Display whether the key is found or not and its position.
   - **Hint**: Depending on the collision resolution strategy (linear probing or chaining), implement the search accordingly.
4. **LEVEL - 2 : Delete Key from Hash Table (Linear Probing)**: Implement a function to delete a key from a hash table that uses linear probing.
   - **Input**: The key to be deleted.
   - **Output**: Display the updated hash table after deletion.
   - **Hint**: Mark the deleted spot in a way that it does not block future probes (use a sentinel value for "deleted").